

BINGLEY, United Kingdom

WMO: 033440

Lat: 53.8113N

Lon: 1.8668W

Elev: 876

StdP: 14.24

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.2	27.4	20.6	15.9	28.5	23.0	17.9	29.7	33.3	43.6	30.3	43.8	6.5	290	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	75.3	63.4	71.5	61.5	68.2	59.9	65.0	71.6	62.9	69.0	61.2	66.3	7.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	87.1	67.7	60.6	81.5	65.1	59.1	77.2	63.2	30.4	71.9	28.8	69.0	27.6	66.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.6	23.0	20.2	DB	21.6	81.1	3.6	4.6	19.0	84.4	16.9	87.1	14.9	89.8	12.2	93.1	(4)
(5)			WB	20.7	69.1	3.4	3.6	18.2	71.7	16.2	73.8	14.3	75.8	11.8	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	38.3	38.4	40.7	44.9	50.2	55.4	59.1	58.5	54.4	49.4	42.9	39.3	(6)	
	DBStd	9.08	5.15	5.44	5.38	5.33	5.40	4.74	4.53	4.23	4.14	4.83	5.14	5.81	(7)	
	HDD50	1847	363	326	291	168	64	7	0	0	7	69	219	333	(8)	
	HDD65	6345	828	745	754	603	460	291	192	208	319	484	664	797	(9)	
	CDD50	997	0	1	2	16	69	168	282	264	139	50	5	1	(10)	
	CDD65	19	0	0	0	0	0	2	9	7	1	0	0	0	(11)	
	CDH74	155	0	0	0	0	3	28	73	47	3	1	0	0	(12)	
	CDH80	15	0	0	0	0	0	1	10	4	0	0	0	0	(13)	
	WSAvg	9.3	11.5	11.6	10.5	9.1	8.7	7.9	7.4	7.5	7.9	9.1	10.1	10.9	(14)	
(15) Precipitation	PrecAvg	40.9	4.4	3.3	2.6	2.6	2.6	2.9	3.1	3.4	3.1	3.9	4.3	4.8	(15)	
	PrecMax	55.6	9.2	8.8	5.4	5.7	5.3	9.2	6.7	8.7	6.4	8.0	9.7	10.7	(16)	
	PrecMin	29.6	0.6	0.7	0.3	0.5	0.6	0.6	1.4	0.6	1.0	1.2	1.7	0.8	(17)	
	PrecStd	7.1	2.0	1.9	1.3	1.4	1.2	2.0	1.4	1.7	1.5	1.7	2.1	2.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	55.0	57.6	66.9	73.1	78.0	80.3	79.7	72.8	64.8	56.8	54.0	(19)	
		MCWB	50.0	49.6	49.7	55.2	62.0	64.1	65.9	63.7	63.9	58.8	55.1	52.0	(20)	
	2%	DB	49.7	50.8	53.5	60.5	68.1	71.9	75.6	73.0	67.2	60.7	54.3	52.0	(21)	
		MCWB	48.0	48.0	48.8	52.6	57.9	60.8	63.6	63.4	59.4	56.7	52.4	50.3	(22)	
	5%	DB	47.7	48.4	50.9	56.6	63.2	67.7	71.8	68.8	63.7	58.5	52.2	49.6	(23)	
		MCWB	46.0	46.2	47.4	50.5	55.3	58.5	61.9	61.2	57.6	55.6	50.3	48.0	(24)	
	10%	DB	46.0	46.4	48.6	53.6	59.4	64.3	67.9	65.8	61.1	56.6	50.3	47.7	(25)	
		MCWB	44.3	44.2	45.7	48.7	52.9	56.7	60.3	59.6	56.4	54.1	48.7	46.0	(26)	
	0.4%	WB	50.8	50.8	51.7	56.9	63.5	65.7	69.3	67.1	64.7	60.3	55.3	52.3	(27)	
		MCDB	51.7	53.3	54.6	63.2	70.9	74.2	76.7	73.1	70.5	62.4	56.6	53.4	(28)	
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.3	48.4	49.5	53.7	59.0	62.0	65.3	64.0	61.3	58.0	52.9	50.5	(29)	
		MCDB	49.4	50.4	52.5	58.9	66.2	69.5	72.3	69.9	65.3	59.8	54.0	51.8	(30)	
	5%	WB	46.4	46.5	47.9	51.4	56.3	59.8	62.9	62.3	59.0	56.2	50.7	48.3	(31)	
		MCDB	47.6	48.2	50.6	56.1	61.5	65.7	69.0	67.6	61.6	58.1	51.8	49.5	(32)	
	10%	WB	44.5	44.4	46.0	49.1	53.9	57.9	61.1	60.7	57.2	54.3	48.9	46.2	(33)	
		MCDB	45.9	46.2	48.4	52.9	58.4	63.1	66.4	64.8	60.1	56.2	50.1	47.5	(34)	
	5% DB	MDBR	7.2	8.3	10.3	12.8	13.7	13.5	13.7	12.6	11.7	9.7	7.7	7.6	(35)	
		MCDBR	9.6	10.2	13.4	17.4	20.3	20.1	20.2	17.5	15.7	11.4	8.9	9.7	(36)	
		MCWBR	9.5	8.7	9.6	10.9	12.5	10.9	10.5	9.2	9.3	8.3	8.4	9.3	(37)	
		MCWBR	9.4	10.0	12.2	16.2	18.2	17.8	17.5	15.4	13.3	10.6	8.9	9.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.314	0.328	0.357	0.382	0.383	0.384	0.384	0.371	0.354	0.338	0.313	0.303	(40)	
		taud	2.322	2.382	2.312	2.262	2.297	2.317	2.317	2.369	2.426	2.444	2.390	2.304	(41)	
	Ebn at Noon		204	244	260	268	274	274	272	269	259	237	205	182	(42)	
		Edh at Noon	19	24	32	38	38	38	37	34	28	22	17	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		200	411	762	1189	1511	1557	1501	1233	892	499	246	155	(44)	
	RadStd		21	47	82	128	102	143	148	81	77	44	26	16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page